Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VII

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Hash Table:

Write a program to:

a. Implement a hash table with separate chaining for collision handling.

Code:

```
size=5
table= [[] for _ in range(size)]

def insert(key,value):
    table[(key%size)].append((key,value))

insert(10,"Mehek")
insert(11,"Nidhi")
insert(12,"Omkar")
insert(13,"Rose")
insert(14,"Kaushal")

print(table)
```

Output:

```
= RESTART: C:/Users/DELL/OneDrive/Desktop/Mehek/SEM 3/
DS/prac7.py
[[(10, 'Mehek')], [(11, 'Nidhi')], [(12, 'Omkar')], [(
13, 'Rose')], [(14, 'Kaushal')]]
>
```

b. Store and retrieve data from the hash table.

Code:

```
size=5
table= [[] for _ in range(size)]

def insert(key,value):
   table[(key%size)].append((key,value))


def search(key):
   for k,v in table[key%size]:
      if k==key:
         return v
   return "Not Found"

insert(10,"Mehek")
insert(11,"Nidhi")
insert(12,"Omkar")
insert(13,"Rose")
insert(14,"Kaushal")

print(table)

key= int(input("Enter key= "))
print(search(key))
```

Output:

```
= RESTART: C:/Users/DELL/OneDrive/Desktop/Mehek/SEM 3/
DS/prac7.py
[[(10, 'Mehek')], [(11, 'Nidhi')], [(12, 'Omkar')], [(
13, 'Rose')], [(14, 'Kaushal')]]
Enter key= 12
Omkar

= RESTART: C:/Users/DELL/OneDrive/Desktop/Mehek/SEM 3/
DS/prac7.py
[[(10, 'Mehek')], [(11, 'Nidhi')], [(12, 'Omkar')], [(
13, 'Rose')], [(14, 'Kaushal')]]
Enter key= 15
Not Found
```